

Documentation part

1. What is an API?

An **API (Application Programming Interface)** is a set of rules that allows different software applications to communicate with each other. It acts as a bridge between a client and a server.

For example, when you use a mobile app, it sends a request to a server through an API and receives a response.

In Machine Learning, APIs are used to send input data to a trained model and receive predictions as output.

2. Difference between Flask and FastAPI

Flask and **FastAPI** are Python frameworks used to build web APIs.

- Flask is a lightweight and simple framework that is easy to learn and widely used.
- FastAPI is a modern framework that is faster and provides automatic data validation and documentation.

Key differences:

- Flask is slower compared to FastAPI.
- FastAPI supports asynchronous programming, Flask has limited support.
- FastAPI automatically generates API documentation, Flask does not.
- FastAPI is better for high-performance applications.

3. What is REST Framework for APIs?

REST (Representational State Transfer) is an architectural style used to design networked applications.

A REST API uses HTTP methods such as:

- GET (retrieve data)
- POST (send data)
- PUT (update data)
- DELETE (remove data)

A REST framework is a tool that helps developers build REST APIs easily. It organizes data into resources and allows communication between client and server in a structured way.

4. What are Microservices and how ML developers use them?

Microservices is an architecture where an application is divided into small, independent services that work together.

Each service performs a specific task, such as:

- Data processing service
- Model prediction service
- Authentication service

In Machine Learning, microservices help separate model training, preprocessing, and prediction into different services. This makes the system easier to scale, maintain, and deploy.

5. What is Docker and how do ML Developers use it?

Docker is a tool used to package applications with all their dependencies into a container so that they can run consistently on any system.

Machine Learning developers use Docker to:

- Deploy ML models easily
- Avoid dependency issues
- Ensure the same environment in development and production
- Share projects with others

6. What is a Dockerfile and why do we create it?

A Dockerfile is a text file that contains instructions to build a Docker image.

It specifies:

- Base operating system
- Required libraries and dependencies
- Application code
- Commands to run the application

We create a Dockerfile to automate the process of setting up the environment and building Docker images.

7. Difference between Docker Image and Docker Container

A **Docker Image** is a read-only template that contains the application and its environment.

A **Docker Container** is a running instance of a Docker image.

Differences:

- Image is static, container is dynamic.
- Image is used to create containers.
- Container is an active running environment.
- One image can create multiple containers.